MILITARY SPECIFICATION
WARHEAD, GUIDED MISSILE,
CHEMICAL AGENT
SMOKE, MK 69 MOD O;
WP FILLING

1. SCOPE

1.1 This specification covers the white phosphorus (WP) filling and closing (without fuzing components) of the Mk 69 Mod O warhead.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-709	-	Ammunition Color Coding.

DRAWINGS

NAVAL ORDNANCE SYSTEMS COMMAND

DL2581064	-	Warhead, Guided Missile, Mk 69 Mod O, Case, Loaded.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:26Z
Check the source to verify that this is the current version before use.

NAVAL AIR SYSTEMS COMMAND

DL1516597 - Container, Shipping and Storage for Mk 396 Mod O (for One Warhead, Guided Missile Mk 52 Mod O), General Arrangement.

DL1516598 - Container, Shipping and Storage for Mk 396 Mod O (for One Warhead, Guided Missile Mk 52 Mod O), Packing and Marking.

US ARMY MUNITIONS COMMAND

C18-60-1293 - Marking, Drawing, Warhead, Guided Missile, Mk 69 Mod O (WP Filled).

(All drawings referenced in this specification are included in, or listed on the above Edgewood Arsenal drawing.)

PUBLICATIONS

BUREAU OF NAVAL WEAPONS

WR-54/105 - Weapons Requirement, Palletizing Fleet Issue Unit Load Mk 72, Mod O.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials and components. All materials and components cited on Drawing 2581064 shall conform to the specifications listed thereon or to the specific characteristics set forth on the drawing. The warhead shall be without fuzing components.

3.2 Leakage. The filled and closed warhead shall not leak agent when maintained within the temperature range 207° to 212°F and tested during a 15-minute period as specified in 4.2.4.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:26Z
Check the source to verify that this is the current version before use.

3.3 Filling.

3.3.1 Filling method. The warhead, clean and free from foreign matter, shall be filled with molten WP by the dry-fill method (see 6.3). Carbon dioxide shall be used to provide a seal against contact with air. After testing for leakage, the WP shall be solidified with the filling port in the up position. (See 6.2.)

3.3.2 Filling height. The filling height shall be such as to provide a 3 to 5 percent void as determined by the method specified on Drawing 2581064 and shall be inspected as specified in 4.2.4.2.

3.4 Closing. The warhead shall be closed as shown on Drawing 2581064 with both balls being pressed simultaneously (see 6.2).

3.5 Cleaning after closing. Any exposed surfaces of the filled and closed warhead which have become contaminated shall be thoroughly cleaned to remove all WP (see 6.4).

3.6 Marking. Marking of the warhead shall be in accordance with Drawing 2581064.

3.7 Workmanship. The warhead shall be free from internal contamination (foreign matter) and damage such as dents, cracks, or deformation which could affect subsequent assembly and functioning and free from external contamination after filling.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:26Z
Check the source to verify that this is the current version before use.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Regular production.

4.2.1 Lotting. A lot shall consist of the filled and closed warheads produced by one manufacturer, at one plant, from the same materials, under essentially the same manufacturing conditions, and offered for delivery at one time.

4.2.2 Sampling. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.3 Inspection.

4.2.3.1 For examination and nondestructive test. Sample warheads shall be examined and tested in accordance with the classification of defects and MIL-STD-105.

4.2.3.2 For critical defects. Each warhead shall be inspected for the critical characteristics listed in the classification of defects.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:26Z
Check the source to verify that this is the current version before use.

4.2.3.3 Classification of defects.

(a) Warhead, guided missile, Mk 69 Mod O, case loaded (Drawing 2581064).

Categories	Defects	Acceptance standards
Critical:		
1	Leakage	4.2.4.1
2	WP contamination (3.5)	
Major:	AQL 0.65 percent defective	
101	Filling height excessive	4.2.4.2
	AQL 1.0 percent defective	
102	Marking incorrect	
103	Damage or internal contamination* (3.7)	
104	Ball seal depth not as specified	
Minor:	AQL 2.5 percent defective	
201	Filling height insufficient	4.2.4.2
202	Marking missing or illegible	
203	Contamination after filling (grease, oil, dirt, or foreign matter) (3.7)	
204	Coating missing, incorrect, or inadequate	

*Internal contamination to be checked prior to filling.

4.2.4 Tests.

4.2.4.1 Leakage. The filled and closed warhead shall be placed in an oven so that the filling port is totally below the level of molten WP and radially in an upper position (and thus subject to view) and observed for leakage while being maintained at the required temperature for the required period of time.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:26Z
Check the source to verify that this is the current version before use.

4.2.4.2 Filling height. The height of WP in the filled and closed warhead shall be radiographed with the warhead in a vertical filling-port-up position and inspected for amount of void.

5. PREPARATION FOR DELIVERY

5.1 Packing, level A. Each filled warhead shall be in a Mk 396, Mod O, Shipping and Storage Container, in accordance with Drawings D1516597 and D1516598.

5.2 Palletization. Filled warheads, packed in accordance with 5.1, shall be palletized in accordance with Weapons Requirement, Palletizing Fleet Issue Unit Load Mk 72, Mod O, WR-54/105.

5.3 Marking.

5.3.1 Warhead. The filled warhead shall be marked with the appropriate nomenclature, as provided in the contract or order. In addition, each warhead shall be provided with ammunition color code marking in accordance with MIL-STD-709, as required for WP smoke.

5.3.2 Shipping container (Mk 396, Mod O). Each shipping container, with warhead intact, shall be marked in accordance with Drawing C18-60-1293.

5.3.3 Palletized load. Each pallet load of warheads shall be provided with a Military Shipping Tag, DD Form 1387-1. The tag shall be marked with the appropriate address/addresses, total weight, and cubic displacement, and other pertinent information specified in the contract or order. At least one container per pallet shall be provided with a Special Handling Data/Certification Label, DD Form 1387-2. Tags and labels shall be securely affixed, the tags by wire or twine and the labels by waterproof adhesive. Marked tags and labels shall be coated with a clear waterproof coating. Marking shall be clear and legible. DD Form 1387-2, when applicable, shall be located on a conspicuous area of the respective container but shall not interfere with other required marking.

6. NOTES

6.1 Ordering data. Procurement documents should specify the title, number, and date of this specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:26Z
Check the source to verify that this is the current version before use.

6.2 Caution. Because of the hazardous nature of WP and dangers inherent in operations employing high pressures, filling installations are advised to take all possible precautions to protect personnel from injury while filling and testing warheads.

6.3 Dry-filling method. It has been determined that the WP vacuum-CO_2 system dry-filling method for smoke munitions described in CRDL Technical Memorandum 61-140 is a satisfactory method for filling the Mk 69 Mod O warhead.

(Copies of CRDL Technical Memorandum 61-140 required by suppliers in connection with specific procurement functions may be obtained from the procuring activity or as directed by the contracting officer.)

6.4 Decontamination. The filled and closed warheads can be decontaminated by scrubbing with hot water or steam and steel wool or wire brush.

Custodian:

Army - MU

Preparing activity:

Army - MU (EA)

Project No. 1336-A124

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:26Z
Check the source to verify that this is the current version before use.

TO DETACH FORM - CUT ON THIS LINE

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 56 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:26Z
Check the source to verify that this is the current version before use.